Dhaivat Naik

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EXPERIENCE

Administrative Analyst 2 – *Information Systems, Department of Community Affairs, The State of New Jersey* ***Jul 2025 – Present***

- Architected a scalable data lakehouse on Databricks over S3 to centralize **11** years of municipal budgeting data (**30M** records) into a medallion architecture, using Airflow to orchestrate ETL triggered by file uploads to S3
- Modeled the silver layer using Data Vault patterns (hubs for municipalities, programs, accounts; links for relationships; satellites for slowly changing attributes) and the gold layer using a star schema (budget and payment fact tables with municipality/program/fiscal period dimensions)
- Implemented event driven ETL in PySpark and Spark SQL orchestrated by Airflow on S3 uploads, reducing the time to assemble standard budget reports from roughly **3+** hours of manual Excel work to under **10** minutes
- Standardized storage in Parquet format and partitioned data by fiscal year and municipality, cutting average query runtime and data scanned for recurring reports by about **~30%** compared to the prior CSV/Excel-based process
- Established Data Governance frameworks via Unity Catalog for automated cataloguing and lineage tracking, implemented fine-grained role-based access column-level protection on PII-adjacent fields

Research Data Engineer, *Rutgers Institute for Health, Healthcare Policy and Aging Research* ***Nov 2023 – Oct 2024***

- Built and maintained a streaming ETL pipeline using Apache Kafka topics (patient-events, vitals, survey-responses) and PySpark; ingesting health data from **10+** sources (wearable sensors, health surveys, NIH datasets) into an Azure Data Lake, processing **~1TB/week** for downstream research
- Designed normalized, patient-centric schemas and star-schema dimensional models, optimized query performance with columnar partitioning and predicate pushdown to achieve **sub-second** query times on datasets with **50M** records

Data Engineering Intern, *SuitUp* ***Jun 2024 – Aug 2024***

- Developed Spark Streaming pipelines on AWS EMR, processing **~1TB/day** of video engagement logs into Redshift tables using Interleaved sort keys and key-based distribution, reducing query scan size by **30%**, for downstream analytics
- Mitigated data skew in workloads by applying **key salting** to high-cardinality (user_id) joins, which optimized **Executor** utilization and prevented Core Node saturation, allowing the pipeline to scale without increasing instance count
- Migrated legacy batch log ingestion process **(3+ hour lag)** to streaming with Amazon Kinesis Data Streams and AWS Glue, achieving **<5 minute** data freshness, enforcing schema through Glue Data Catalogue

Data Scientist, *SAS Institute* ***Jan 2022 – Jul 2023***

- Developed automated feature pipelines for BFSI clients (credit risk, customer churn, behavioural segmentation, cross-sell, attrition) translating business requirements and raw financial data into analytical datasets on SAS Viya contributing to **$1.2M** in closed deals
- Collaborated with account executives, solution architects, and client stakeholders to respond to RFPs, define success metrics for ML solutions and design technical data architectures, contributing to POC-to-contract conversion
- Defined and enforced data SLAs for model inputs (market open batch refresh and sub hourly updates for near real time scoring), adding freshness and data completeness checks that cut model failures in client environments by roughly **25%**
- Architected data governance by cataloging critical datasets and features, and documented lineage from source systems into analytical tables, implementing Role Based Access Control for sensitive PII attributes
- Created **20+** interactive dashboards in SAS Visual Analytics and PowerBI, showcasing model predictions and simulations for client demos and pre-sales POCs

CERTIFICATIONS

AWS Data Engineer – Associate [DEA-C01]

Databricks Certified Associate Developer for Apache Spark

PROJECTS

FitFindr – Multi-Tool AI Agent: Built a stateful AI agent in Python using Groq, Gradio and structured JSON listing data that parsed natural-language fashion queries into search filters to retrieve relevant thrift items. Implemented a planning loop that chose the next tool based on prior results, reused intermediate outputs through shared session dict, implemented retry and fallback logic, and used pytest to validate main error paths

Spark RAG Assistant – Built a Python-based RAG pipeline using sentence-transformer (MiniLM), Chroma vector store, and Groq LLM that ingested and chunked official Spark documentation into a semantic index, orchestrated retrieval of top-k chunks per query, enforced grounded generation via a system prompt, maintained state across CLI and Gradio interfase and documented evaluation, failure cases and error handling

SKILLS

- **Languages:** Python, SQL, PySpark, Scala
- **Cloud & Infrastructure:** Azure Data Lake, AWS Redshift, S3, Lambda, SQS, Athena, Snowflake, Databricks
- **Databases:** PostgreSQL, MySQL, MongoDB, Cassandra, DynamoDB
- **Data Engineering Stack:** Apache (Spark, Kafka, Airflow), dbt, Docker, Hadoop, Kubernetes
- **Machine Learning Libraries:** Pytorch, Sci-kit learn, NLTK, Numpy, Pandas, Huggingface, XGBoost,
- **Visualization:** PowerBI, Tableau, Quicksight, SAS Visual Analytics, Matplotlib

EDUCATION

- Rutgers University – **Master of Science in Data Science – Statistics**
- Narsee Monjee Institute of Management Studies (NMIMS) – **Bachelor of Technology in Computer Engineering**